3rd partial exam of Combinatorics & Probability **Univ. Rovira i Virgili**
January 9, 2022 **Version B**

You have 2 hours for completing the exam.
No electronic devices are allowed (i.e. calculators, mobile phones, etc.).
All answers must include the supporting deductions and calculations (if the calculations are too heavy then they can be left implied in the final answer).
If some answer does not include all the necessary information, it will be deemed invalid.

Name and surname:

1. Let χ be a discrete random variable that takes positive integer values with probabilities
$$p_k = \mathbb{P}(\chi = k) = M\left(\frac{4}{5}\right)^k,$$
for $k = 1, 2, 3, \ldots$ and for some constant $M \in \mathbb{R}$.
   (a) (1 point) Determine the value of M.
   (b) (1 point) Find the Probability Generating Function (PGF) associated with the sequence $\langle p_k \rangle_{k=1}^{\infty}$.
   (c) (1 point) Calculate the expected value $\mathbb{E}(\chi)$ and the variance $\mathbb{V}(\chi)$.

2. Let χ be a discrete random variable with the following PMF:
$$p_\chi(k) = \begin{cases} \frac{1}{3} \text{ for } k = 0 \\ \frac{1}{6} \text{ for } k = 1, -1, 2, -2 \\ 0 \text{ otherwise.} \end{cases}$$
   (a) (1 point) We define a new random variable $\psi = (\chi - 1)^2$. Construct the PMF of ψ.
   (b) (1 point) Compute the expected value and the variance of ψ.
   (c) (1 point) Calculate $\mathbb{P}(\psi \geq 4)$ and $\mathbb{P}(\psi \leq 1)$.

3. Let χ be a continuous random variable with Probability Density Function (PDF)
$$f(x) = \begin{cases} 0 \;\text{ for } x \in (-\infty; 1) \\ \frac{c}{x^5} \;\text{ for } x \in [1; \infty) \end{cases}$$

(a) (1 point) Determine the value of c and the Cumulative Distribution Function (CDF) $F(x)$ of χ.

(b) (1 point) Compute the expected value of χ (if it exists).

(c) (2 points) Compute the variance of χ (if it exists).

SOME USEFUL TABLES AND FORMULAS

$$\mathbb{E}(\chi) = \sum_{s \in S} s\, p_\chi(s), \qquad \mathbb{E}(\chi) = \int_{-\infty}^{\infty} t\, f_\chi(t)\, dt$$

$$\mathbb{V}(\chi) = \mathbb{E}\left((\chi - \mathbb{E}(\chi))^2\right), \qquad \mathbb{V}(\chi) = \mathbb{E}\left(\chi^2\right) - \mathbb{E}(\chi)^2$$

$$\mathbb{V}(\chi) = \sum_{s \in S} (s-\mu)^2\ p_\chi(s), \quad \mathbb{V}(\chi) = \int_{-\infty}^{\infty} (t-\mu)^2\ f_\chi(t)\, dt$$

$$\mathbb{E}(\chi) = P'_\chi(1), \qquad \mathbb{V}(\chi) = P''_\chi(1) + P'_\chi(1) - [P'_\chi(1)]^2$$

Table of ordinary generating functions

Function	Sequence	Series
$\frac{1}{1-az}$	$\langle 1, a, a^2, a^3, \ldots \rangle$	$\sum_{n=0}^{\infty} a^n z^n$
$\frac{1}{1-z^2}$	$\langle 1, 0, 1, 0, 1, \ldots \rangle$	$\sum_{n=0}^{\infty} z^{2n}$
$\frac{1}{(1-z)^2}$	$\langle 1, 2, 3, 4, 5, \ldots \rangle$	$\sum_{n=0}^{\infty} (n+1) z^n$

Discrete probability distributions

Distribution	p_k (PMF)	Mean	Variance	PGF
Discrete uniform	$\frac{1}{n}$	$\frac{a+b}{2}$	$\frac{n^2-1}{12}$	$\frac{z^a - z^{b+1}}{n(1-z)}$
Binomial	$\binom{n}{k} p^k q^{n-k}$	np	npq	$(q+pz)^n$
Geometric	pq^{k-1}	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pz}{1-qz}$
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	$e^{\lambda(z-1)}$